

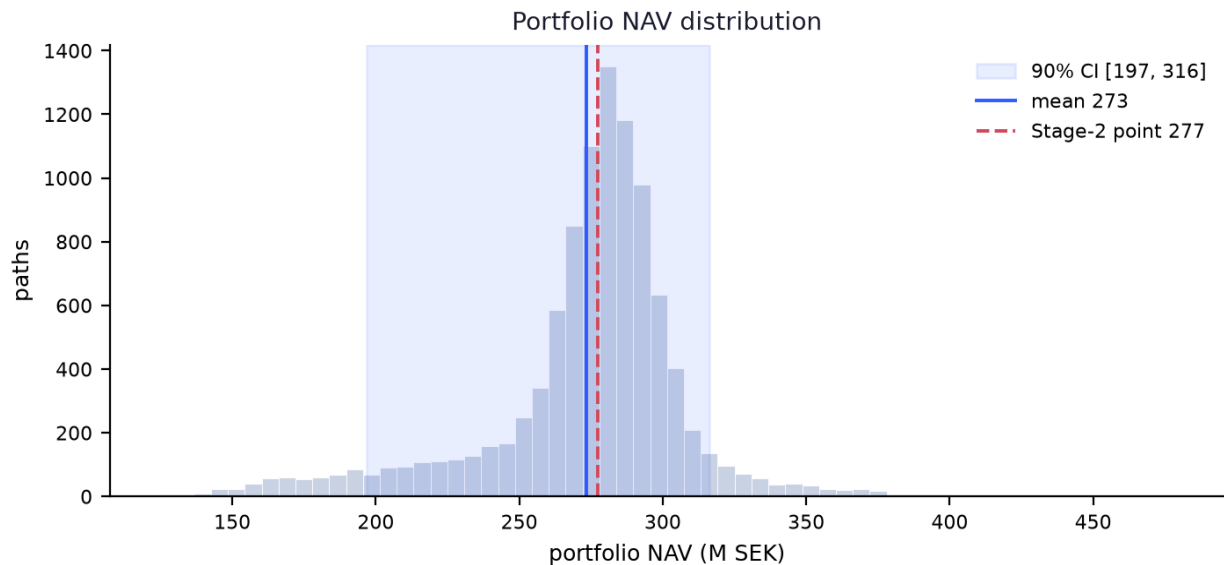
# PVE — Stage 3: Uncertainty Quantification

NAV as a distribution · correlated Monte Carlo over valuation inputs · repo ~/dev/pve · 2026-07-05

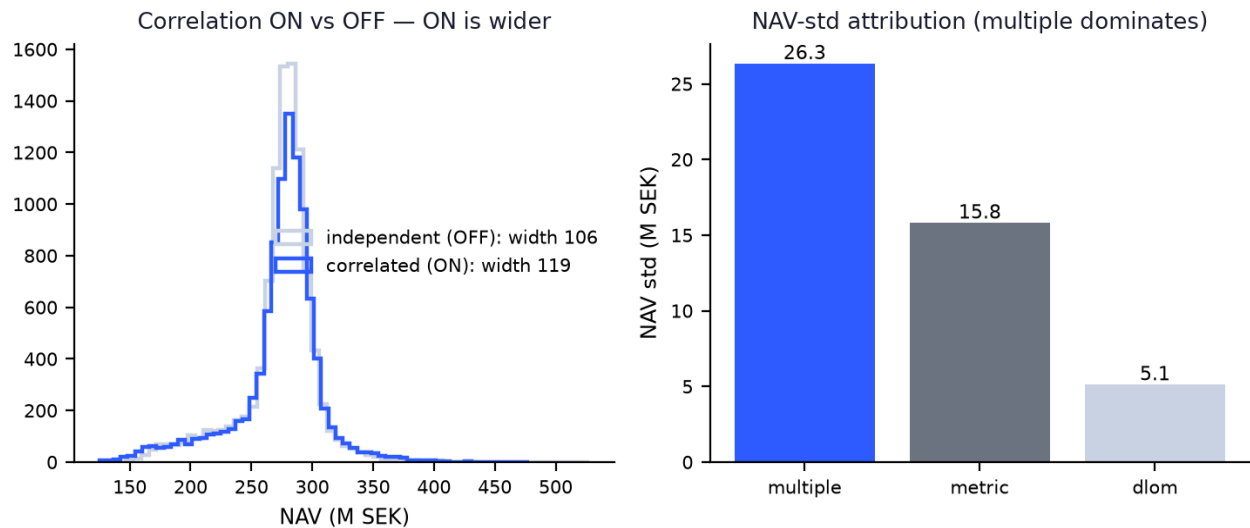
|                                                   |                                                    |                                                  |                                            |
|---------------------------------------------------|----------------------------------------------------|--------------------------------------------------|--------------------------------------------|
| <b>Portfolio NAV</b><br>273 M · 90% CI [197, 316] | <b>Correlation</b><br>ON band 1.13× wider than OFF | <b>Convergence</b><br>bounds stable n→2n (<0.03) | <b>Dominant driver</b><br>implied multiple |
|---------------------------------------------------|----------------------------------------------------|--------------------------------------------------|--------------------------------------------|

## From a number to a number with honest uncertainty

Stage 2 asserted NAV = 277 M. Stage 3 states NAV = **273 M with a 90% credible interval of [197, 316]**. Each input (implied multiple, operating metric, DLOM) carries a distribution; draws are pushed through the exact Stage-2 payoff curve via evaluate\_array. The Stage-2 point mark sits at the centre of the distribution (dashed red).



## The correlation requirement, and what drives the uncertainty



**Left:** multiples move together (common + sector factors), so the correlated NAV band (119 M) is 1.13× wider than the independent one (106 M) — drawing holdings independently would fake diversification and understate risk. **Right:** the implied multiple dominates portfolio variance (std 26.3 vs metric 15.8 vs DLOM 5.1), as expected.

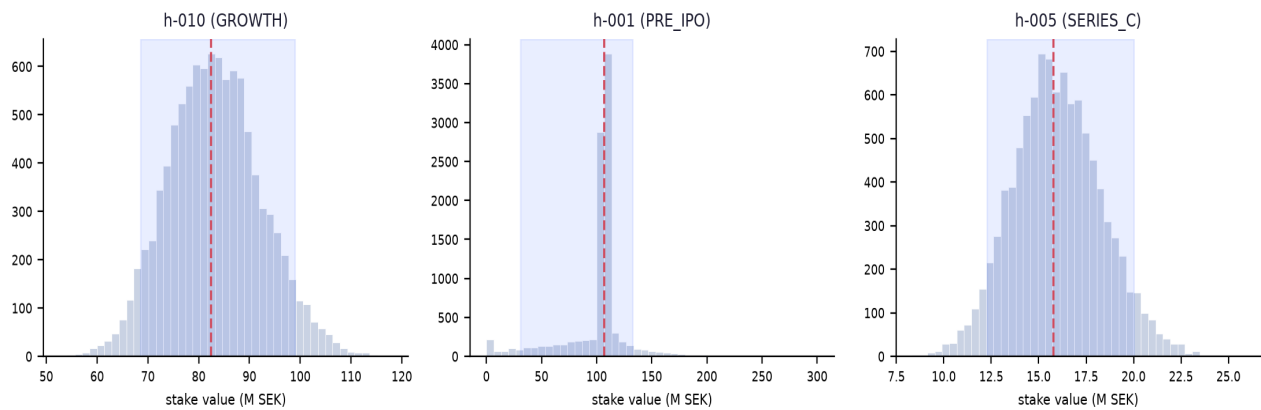
## Gap decomposition — why each flagged name is flagged

| holding | self   | indep | gap%   | comps | dlom   | driver | flag       |
|---------|--------|-------|--------|-------|--------|--------|------------|
| h-006   | 24.32  | 14.28 | -41.3% | -5.13 | -4.92  | comps  | OVERMARKED |
| h-010   | 100.79 | 82.39 | -18.2% | +9.98 | -28.37 | dlom   | OVERMARKED |
| h-000   | 0.72   | 0.61  | -16.2% | +0.83 | -0.95  | dlom   | OVERMARKED |

| holding | self   | indep  | gap%   | comps  | dlom   | driver | flag         |
|---------|--------|--------|--------|--------|--------|--------|--------------|
| h-005   | 18.60  | 15.76  | -15.2% | +5.98  | -8.82  | dlom   | OVERMARKED   |
| h-002   | 0.43   | 0.36   | -15.2% | +0.50  | -0.57  | dlom   | OVERMARKED   |
| h-008   | 18.78  | 21.22  | +13.0% | +9.76  | -7.31  | comps  | conservative |
| h-003   | 4.61   | 4.13   | -10.6% | +1.82  | -2.31  | dlom   | OVERMARKED   |
| h-004   | 7.32   | 6.94   | -5.2%  | +3.50  | -3.88  | dlom   |              |
| h-001   | 102.66 | 106.63 | +3.9%  | +24.85 | -20.88 | comps  |              |
| h-009   | 22.98  | 22.58  | -1.7%  | +4.03  | -4.42  | dlom   |              |
| h-007   | 0.36   | 0.37   | +1.7%  | +0.59  | -0.58  | comps  |              |
| h-011   | 2.16   | 2.12   | -1.6%  | +1.67  | -1.70  | dlom   |              |

The two components are additive to the total gap (reconciles to 1e-9). A name flagged because the **comps disagree** (comps-driven) is a stronger signal than one flagged merely for a large DLOM.

## Per-holding value distributions (Stage-2 point mark dashed)



Convergence: 90% bounds move <0.017 from 10000 to 20000 paths (NAV mean SE 0.35). Bands are wider for earlier-stage and None-metric holdings (more input uncertainty); every interval contains its point mark and no draw is negative.

## Judgment calls / TODO

**DLOM central = Chaffe** (middle method) so the [empirical, Finnerty] range is symmetric around it and the point mark sits at the distribution centre; drift detection is invariant to the DLOM level. **Metric/DLOM shocks idiosyncratic**, multiples correlated (the material choice). **TODO(stage-4):** exit timing / forward paths (Stage 3 is uncertainty on *today's* value only). **TODO(stage-5):** the common/sector factor structure here should feed the risk model's covariance for consistency.